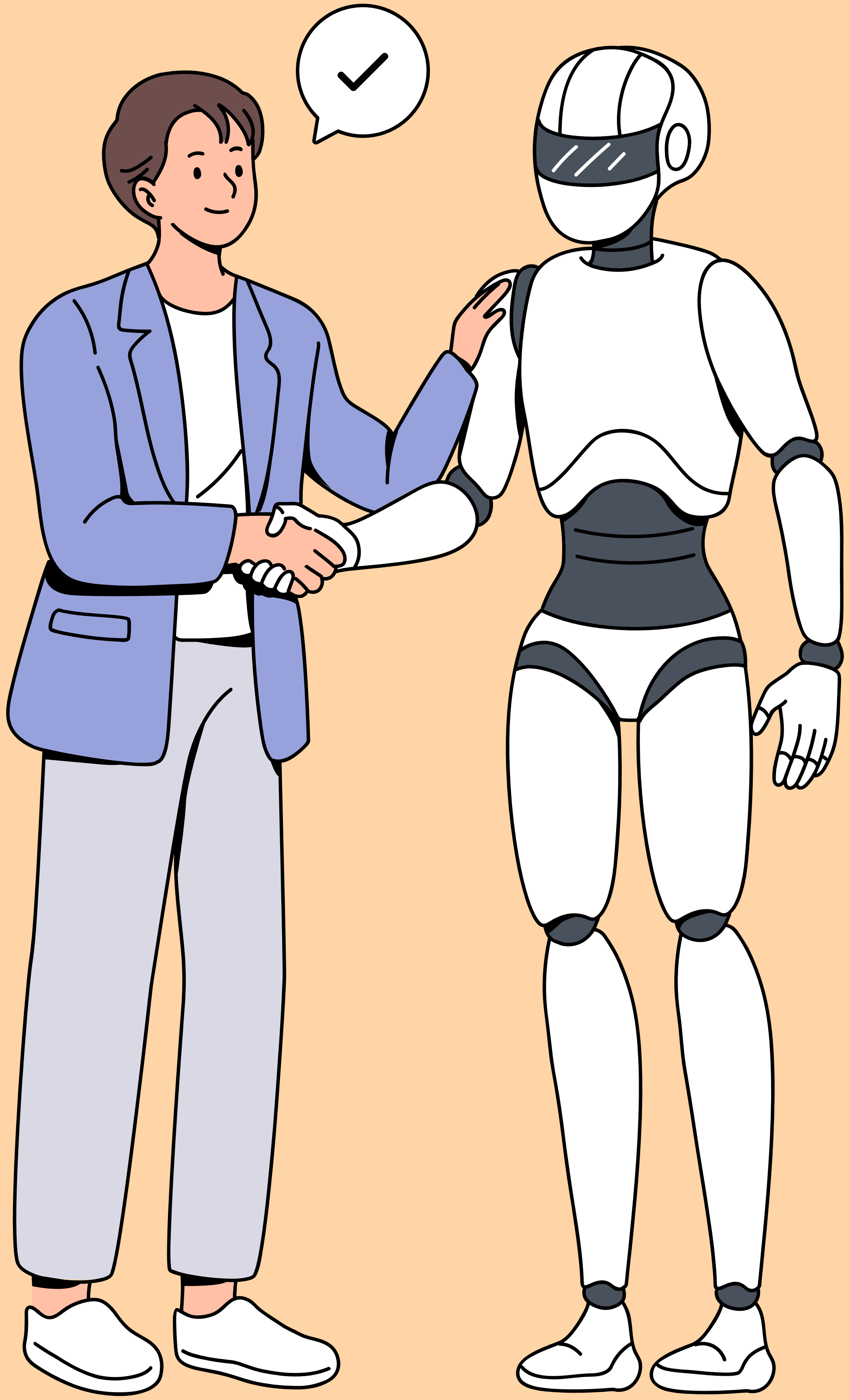
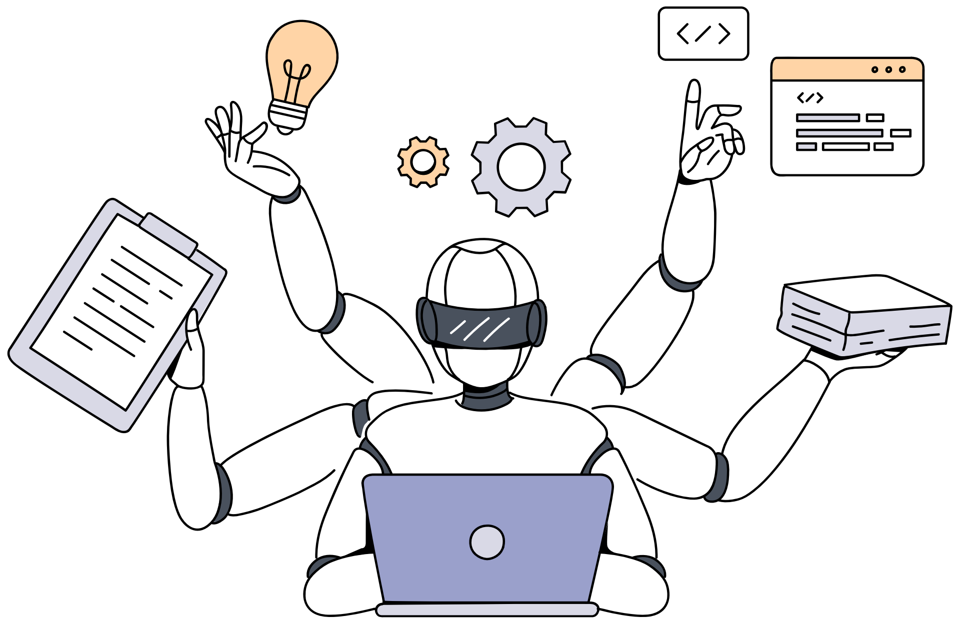


YILDIZ TEKNİK ÜNİVERSİTESİ
BİLGİSAYAR MÜHENDİSLİĞİ
BLM3510- Yapay Zeka
Grup 2- Öğr.Gör.Dr. AHMET ELBİR
2. Odev Reporu
vedio Link:
https://youtu.be/pyNVQ4GeOTQ

Drive link:
https://drive.google.com/drive/folders/16C0zk
mfOy6m9HANR54DhFi5m0AqCryA4?
usp=sharing

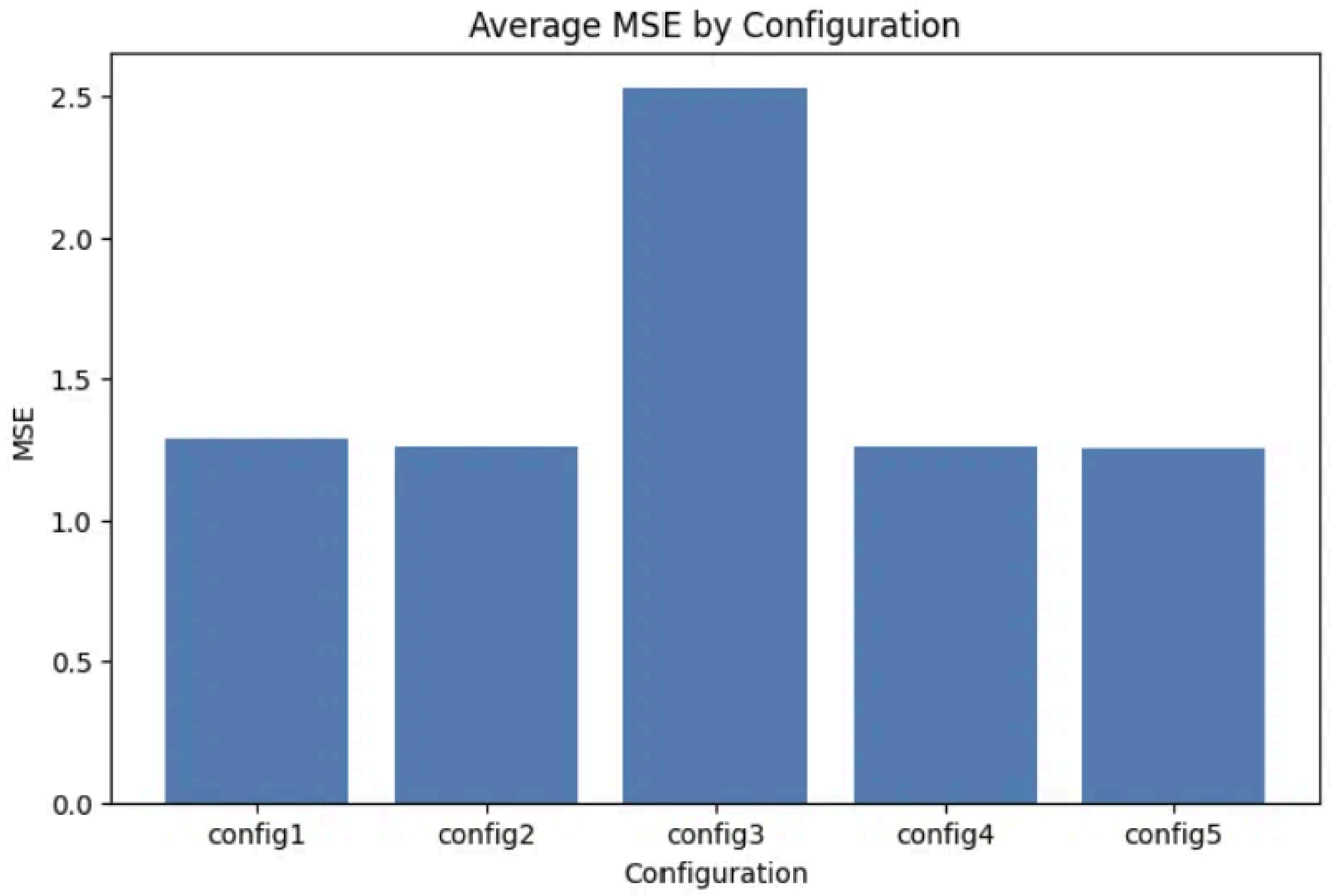
OUTERVEIW

The goal of this HW was to experiment with diffrent LLM models, configurtrations and algorithms to compare them and find thei results. A program was trained to predect the scores given by humans from 1-5 in a report. The dataset was really large and took lots of time to process. Each row consisted of a question (S), a reasoning/thinking process (D), and an answer (C). The data was split into a training set and a test set of 1,000 randomly selected rows using a fixed random seed to correctly compare the models and the algorithms. 5 configurations were created and saved in a drive folder: S only, D only, C only, S+D combined, and D+C combined. Using 4 models from HuggingFace: ytu-ce-cosmos/turkish-e5-large, jinaai/jina-embeddings-v5-text-small, microsoft/harrier-oss-v1-0.6b, and Qwen/Qwen3-Embedding-0.6B, each one was converted into embeddings to easly use it in the algorithms later. Due to the large amout of space required google colab and Kaggle were used.
Three regression algorithms were then trained on each embedding set: Linear Regression, Ridge Regression, and Lasso Regression. This produced a total of 60 experiments (5 configurations × 4 embedding models × 3 algorithms). Each experiment was then evaluated using Mean Squared Error (MSE) and R^2 score on the test set. The results are shown in this report.

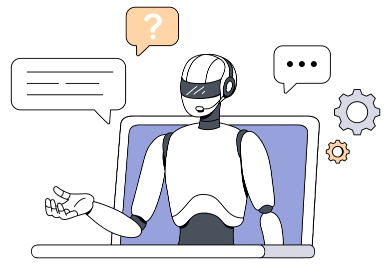


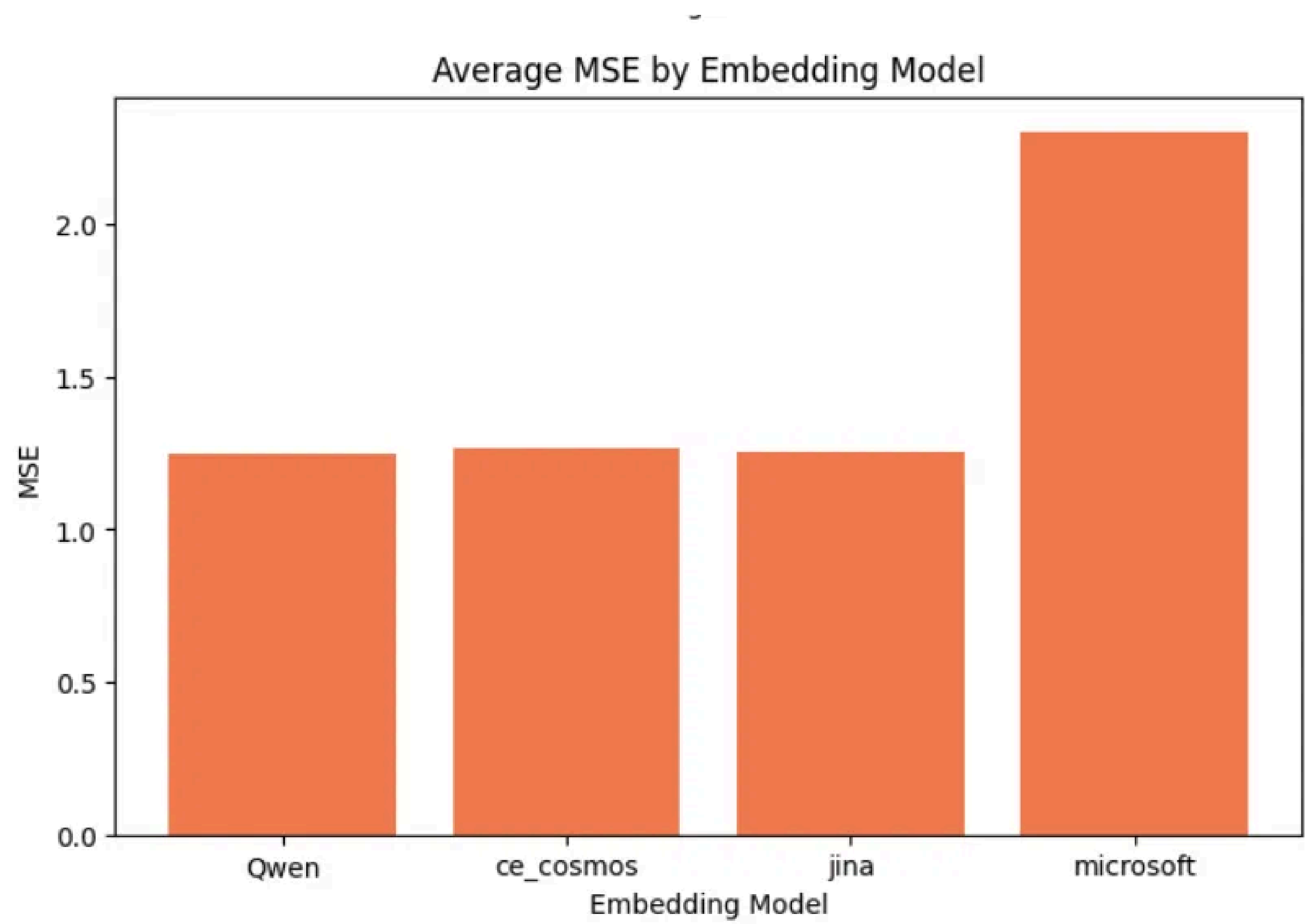
BY CONFIGURATION

Config3 (the answer text only) performed the worse out of all the others with MSE about 2.5. Configs 2, 4, and 5 performed very similarly. This means that the answer text alone is not useful for predicting scores. Looking at the question only Config1 gave the best results



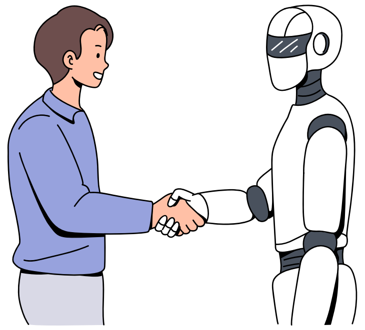
config	MSE	R2
config1	1.2903	-0.0231
config2	1.2598	0.0010
config3	2.5294	-1.0053
config4	1.2597	0.0012
config5	1.2509	0.0081





embedding_model	MSE	R2
Qwen	1.2507	0.0083
ce_cosmos	1.2649	-0.0030
jina	1.2541	0.0056
microsoft	2.3023	-0.8254

MSE R2

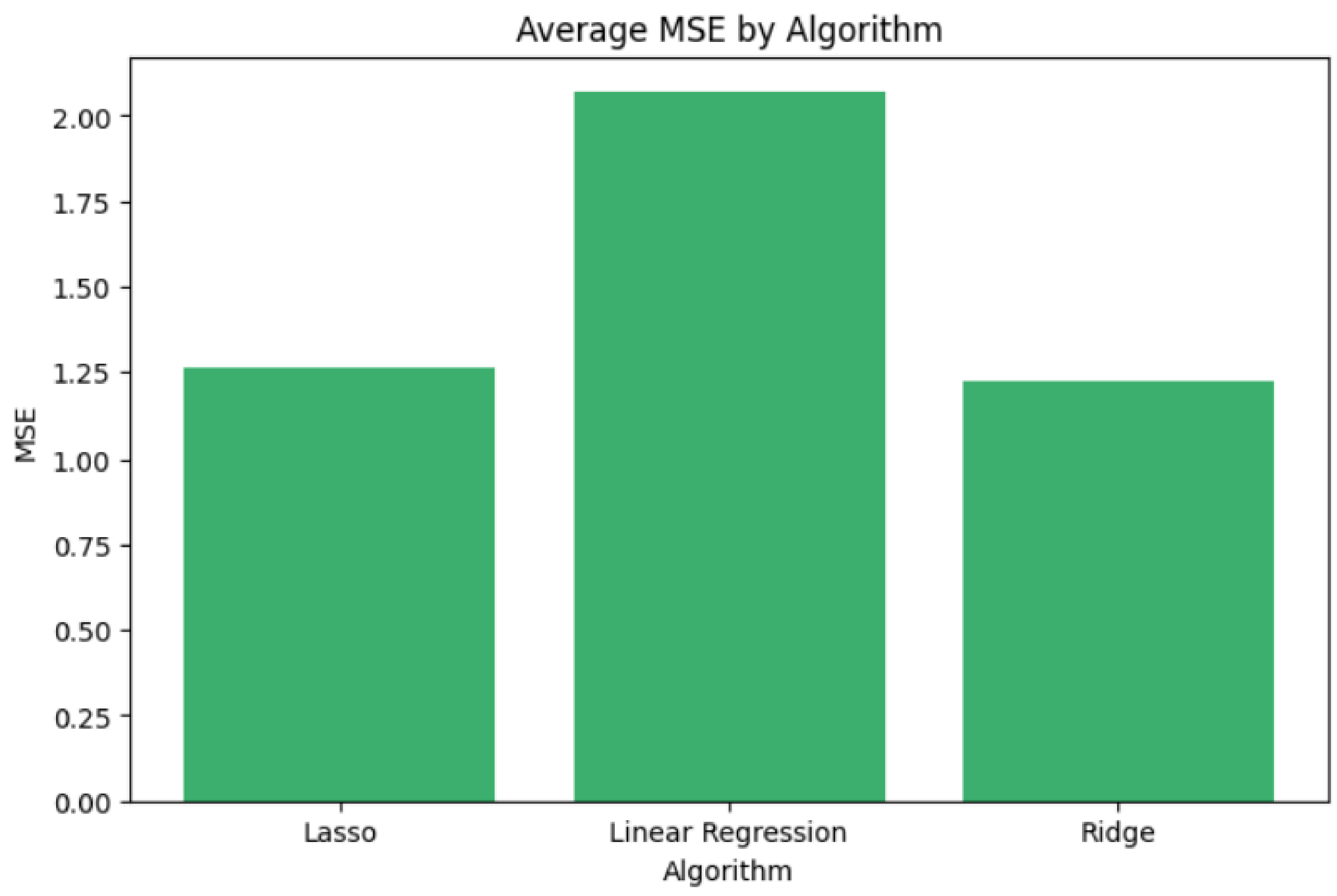


BY MODEL

Qwen, ce_cosmos, and Jina all performed similarly (about 1.25 MSE). Microsoft performed the worst (about 2.3 MSE).

BY ALGORITHM

Ridge was the best, Lasso second, and Linear Regression worst. This makes sense because Ridge adds a small penalty that prevents overfitting on high-dimensional embedding vectors.



algorithm	MSE	R2
Lasso	1.2621	-0.0008
Linear Regression	2.0699	-0.6411
Ridge	1.2220	0.0310